# Egg Drops with Four Eggs

A corrected 1-3 versus 2-2 comparison: the first egg is dropped at nonuniform, optimal floors

**Central correction**

The 1-3 strategy does **not** use a fixed interval and does **not** introduce superfloors. The first egg is placed at a sequence of optimally chosen floors. If it breaks, the remaining uncertainty is solved by the genuine three-egg problem. The resulting variable-gap schedule is exactly the standard four-egg dynamic strategy, and its leading constant is 24^(1/4), not 4(2/9)^(1/4).


## Abstract

This paper revisits the four-egg comparison in the supplied lecture and paper, correcting the interpretation of the 1-3 strategy. The source materials distinguish dynamic search from fixed-step search and record the classical capacity formula: with k eggs and d drops, one can resolve at most sum_{j=1}^k C(d,j) floors. We use that result to write the 1-3 strategy explicitly. The first egg is dropped at a sequence of floors with decreasing increments; the increment at stage t is one plus the number of floors that three eggs can resolve with the remaining drop budget. This is not a superfloor construction. It yields a total capacity asymptotic to d^4/24, hence a worst-case drop count of (24N)^(1/4) + O(1). We then compare this with the fixed-block 2-2 construction, whose leading term is 2 sqrt(2) N^(1/4).

For large N, the leading constants are approximately 2.21336 for 1-3 and 2.82843 for 2-2. The key mathematical point is therefore not a small correction to the previous calculation: once the first egg is allowed to move to optimal nonuniform floors, the structure of the 1-3 strategy changes qualitatively.


## 1. The egg-drop model

There are N floors and four identical eggs. There is an unknown breaking floor b: an egg dropped from a floor below b survives, while an egg dropped from floor b or higher breaks. The goal is to determine b in the fewest drops in the worst case. A broken egg is lost; a surviving egg can be reused.

The distinction between *dynamic* and *fixed-step* search matters. The supplied Fibonacci Quarterly paper explicitly notes that the classical binomial-capacity result uses dynamic search, where a drop decision depends on previous outcomes, while the paper's main one-dimensional theorem analyzes a simpler fixed-step strategy. The present 1-3 analysis uses the dynamic viewpoint because the first egg's floors are not equally spaced.

## 2. The capacity viewpoint

It is easier to count how many floors can be guaranteed with a fixed drop budget than to minimize drops directly. Let A_k(d) be the maximum number of floors that can be handled with k eggs and at most d drops. The classical recurrence is

$$A_k(d) = 1 + A_{(k-1)(d-1)} + A_k(d-1), \quad A_k(0) = 0.$$

The three terms correspond to the floor tested now, the region below it if the egg breaks, and the region above it if the egg survives. Iterating the recurrence gives

$$A_k(d) = C(d,1) + C(d,2) + \ldots + C(d,k).$$

This is the binomial-sum formula cited in the supplied paper from Boardman's work. The important feature for four eggs is that A_4(d) grows like d^4/24, while A_3(d) grows like d^3/6.

$$A_3(d) = d + C(d,2) + C(d,3) = d(d^2 + 5)/6.$$

$$A_4(d) = d + C(d,2) + C(d,3) + C(d,4) = d(d+1)(d^2 - 3d + 14)/24.$$

## 3. The corrected 1-3 strategy

The phrase '1-3' describes the resource split after the first egg eventually breaks: the first egg has been used for a sequence of screening drops, and the remaining three eggs solve the residual interval. Crucially, the first egg is **not** dropped every F floors. Instead, its next floor is chosen from the amount of uncertainty that can still be resolved by the three remaining eggs.

To make the construction exact, fix a total drop budget d. Suppose the first egg has survived t-1 screening drops. There are d-t drops left after the next drop. If the next first-egg drop breaks, the lower uncertain region must therefore contain at most A_3(d-t) floors, because that is exactly what three eggs can resolve in d-t further drops. Hence the next increment in the first-egg schedule should be

$$gap_t = A_3(d-t) + 1.$$

The '+1' is the breaking floor itself: once the first egg breaks there, that floor is known to be a breaking floor, so the three-egg subproblem concerns the floors below it.

Thus the first egg is dropped at nonuniform floors x_1, x_2, ..., where

$$x_0 = 0, \quad x_t = x_(t-1) + A_3(d-t) + 1.$$

As t increases, the remaining drop budget decreases, so A_3(d-t) decreases. The jumps therefore shrink. This is the four-egg analogue of the familiar two-egg triangular-number strategy discussed in the talk, where the increments go x, x-1, x-2, and so on.

## 4. Why the schedule is optimal

There is a particularly clean way to see the result. Summing the variable gaps gives the total number of floors covered after d drops:

$$N_max(d) = sum_(t=1)^d [A_3(d-t) + 1].$$

Re-indexing with r = d-t gives

$$N_max(d) = d + sum_(r=0)^(d-1) A_3(r).$$

Now use the hockey-stick identity sum_{r=0}^{d-1} C(r,j) = C(d,j+1). Since A_3(r) = C(r,1)+C(r,2)+C(r,3),

$$N_max(d) = C(d,1)+C(d,2)+C(d,3)+C(d,4) = A_4(d).$$

**Interpretation**

The corrected 1-3 strategy fills exactly the same capacity A_4(d) as the optimal dynamic four-egg problem. So the variable-gap first-egg construction is not merely a better constant-factor approximation. Under the stated 1-3 architecture, it recovers the standard four-egg optimum.

## 5. An explicit example: eight drops

With d = 8 total drops, A_4(8) = 162 floors. The first egg can therefore be dropped at the following floors. The third column is the jump size; the fourth column is the number of lower floors that three eggs can still resolve if the first egg breaks on that drop.

| Drop t | First-egg floor x_t | Gap | 3-egg capacity after break |
|---|---|---|---|
| 1 | 64 | 64 | 63 |
| 2 | 106 | 42 | 41 |
| 3 | 132 | 26 | 25 |
| 4 | 147 | 15 | 14 |
| 5 | 155 | 8 | 7 |
| 6 | 159 | 4 | 3 |
| 7 | 161 | 2 | 1 |
| 8 | 162 | 1 | 0 |

The gaps are 64, 42, 26, 15, 8, 4, 2, 1. They are emphatically not constant. If the first egg breaks on floor 106, for example, the lower uncertain interval has 41 floors, and three eggs with the remaining six drops can resolve exactly A_3(6) = 41 floors. Every branch is therefore budgeted to fit.

## 6. Turning capacity into a drop-count asymptotic

Let D_k(N) denote the minimum number of drops needed in the worst case with k eggs and N floors. From the capacity formula, D_4(N) is the least d for which A_4(d) >= N. Since

```
A_4(d) = d^4/24 + O(d^3),
```

inverting the leading term gives

```
D_4(N) = (24N)^(1/4) + O(1).
```

Numerically, the leading coefficient is

```
24^(1/4) = 2.2133638394...
```

This is the corrected leading constant for the 1-3 strategy when its first egg is placed at optimally chosen nonuniform floors and the resulting three-egg subproblem is solved optimally.

## 7. The 2-2 strategy really does use superfloors

The comparison strategy is structurally different. Here we partition the building into blocks of F floors, called superfloors. The first two eggs solve a two-egg problem on approximately N/F superfloors; once the critical superfloor is identified, the remaining two eggs solve the two-egg problem inside that block.

For two eggs, the exact capacity after d drops is

```
A_2(d) = C(d,1) + C(d,2) = d(d+1)/2.
```

Thus the exact two-egg drop count for m floors is

```
D_2(m) = ceil((sqrt(8m+1) - 1)/2).
```

Ignoring only integer-rounding effects, the 2-2 total is therefore

```
T_22(N,F) = sqrt(2N/F) + sqrt(2F) + O(1).
```

This is where a calculus optimization is appropriate: F is genuinely a fixed block size in this strategy.

## 8. Optimizing the 2-2 block size

Differentiate the continuous approximation:

```
T_22'(F) = -(sqrt(2N)/2) F^(-3/2) + (sqrt(2)/2) F^(-1/2).
```

Setting the derivative to zero gives

```
F = sqrt(N).
```

Substituting this into the total gives

```
T_22(N) = 2 sqrt(2) N^(1/4) + O(1).
```

Hence the leading coefficient for 2-2 is

```
2 sqrt(2) = 2.8284271247...
```

## 9. Side-by-side comparison

| Strategy | Structure | Leading term | Constant |
|---|---|---|---|
| 1-3 | One first egg; nonuniform optimal floors; after a break, solve the remaining interval with 3 eggs | (24N)^(1/4) | 2.213363839 |
| 2-2 | Fixed blocks of F floors; solve superfloor problem with 2 eggs, then block with 2 eggs | 2 sqrt(2) N^(1/4) | 2.828427125 |

The ratio of leading constants is 0.782542. Thus the 1-3 leading constant is about 21.7% smaller than the 2-2 leading constant. This comparison concerns the asymptotic worst-case drop count as N becomes large; finite-N rounding can make the exact integer drop counts closer for particular buildings.

## 10. Finite-N check

The following values use the exact binomial capacity for 1-3 and the exact two-egg formula together with an integer search over F for 2-2.

| N | 1-3 exact drops | 2-2 exact drops | 2-2 block F | 2-2 split |
|---|---|---|---|---|
| 10,000 | 23 | 28 | 66 | 17 + 11 |
| 1,000,000 | 71 | 89 | 817 | 49 + 40 |
| 100,000,000 | 222 | 282 | 9,448 | 145 + 137 |

For N = 100,000,000, for example, the exact 1-3 count is 222 drops because A_4(221) < 100,000,000 <= A_4(222). A 2-2 search optimized over integer F can be done in 282 drops with F = 9,448 (145 drops on the superfloor problem and 137 inside the final block).

## 11. Why the earlier calculation was not the 1-3 strategy

A fixed-interval calculation starts by choosing a single S and then dropping the first egg at S, 2S, 3S, .... That construction has a worst-case first stage of about N/S drops. It can be useful as a simple fixed-step upper bound, and the supplied paper analyzes such fixed-step strategies. But it is not the strategy described here.

In the corrected 1-3 strategy, the first-egg spacing is allowed to change after every surviving drop. The reason is structural: after t-1 successful drops, there are fewer drops left for the eventual three-egg search. The remaining three-egg capacity therefore falls from A_3(d-1) to A_3(d-2), then to A_3(d-3), and so on. The correct next gap tracks those changing capacities.

> **The essential distinction**
>
> **1-3:** variable first-egg floors, no superfloor parameter; after break, three-egg subproblem.
> **2-2:** fixed block size F, superfloor problem with two eggs, then a two-egg problem inside the block.

## 12. Relation to the supplied talk and paper

The July 28, 2026 talk explicitly introduces the two-egg improvement by saying not to move in constant steps: the increments decrease dynamically. It then says, for three eggs, that once one egg cracks, the problem becomes the two-egg problem. The slides later contrast a 1-versus-3 split with a 2-versus-2 split. The Fibonacci Quarterly paper likewise distinguishes dynamic search from fixed-step strategies and records the binomial capacity formula.

The present paper takes the dynamic capacity statement from those materials and specializes it to the four-egg 1-3 architecture. The exact nonuniform floor sequence and the equality N_max(d) = A_4(d) are the mathematical inference that makes the correction precise.

## 13. Final conclusion

For four eggs, the two strategies do not have the same leading constant. Once 1-3 is interpreted correctly, there is no fixed interval S at all. The first egg is dropped at a sequence of optimally chosen floors whose gaps are determined by the remaining three-egg capacity. In exact capacity language, the schedule satisfies

```
gap_t = A_3(d-t) + 1,    sum_t gap_t = A_4(d).
```

Consequently, the 1-3 strategy requires

```
D_13(N) = (24N)^(1/4) + O(1),    leading constant 2.2133638394...
```

By contrast, the fixed-superfloor 2-2 construction requires

```
D_22(N) = 2 sqrt(2) N^(1/4) + O(1),    leading constant
                      2.8284271247...
```

The corrected analysis therefore makes the intended distinction sharp: 1-3 is a dynamic, nonuniform first-egg schedule; 2-2 is the fixed-block superfloor strategy.

## References

[1] X. Cao, Z. Chen, and S. J. Miller, *Egg Drop Problems: They Are All They Are Cracked Up to Be!*, Fibonacci Quarterly, compiled May 10, 2026. The supplied paper; see especially the discussion of dynamic versus fixed-step search and the statement of the binomial-capacity formula.

[2] S. J. Miller, *Egg Drop Mathematics: It IS all its cracked up to be!*, Hampshire lecture slides, July 28, 2026. The talk's two-egg dynamic rescaling slides and the later 1-3 / 2-2 discussion motivate the strategy distinction used here.

[3] M. Boardman, 'Egg drop numbers,' *Mathematics Magazine* 77(5), 368-372 (2004). Referenced by the supplied paper for the binomial-sum capacity formula.

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